Phase Space Debug Report: 5_1^2 (SnapPy: `L5a1`)

Automated Pipeline

March 8, 2026

1 Gluing Equations (SnapPy)

Manifold: `L5a1`, $n = 4$ tetrahedra, $r = 2$ cusps. Columns ordered as $Z_1, Z_1', Z_1'',\ Z_2, Z_2', Z_2'',\ Z_3, Z_3', Z_3'',\ Z_4$

Tetrahedra equations (C matrix, shape 4×12):

$$C = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

Cusp equations (row order: μ_0, λ_0, μ_1, λ_1):

$$C_{\text{cusp}} = \begin{bmatrix} 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & -1 \\ -1 & 1 & 1 & 0 & -2 & 0 & -1 & 1 & 1 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & -1 & 0 \end{bmatrix}$$

2 Geometric Basis

The pipeline finds $n - r = 2$ independent internal edges. Both are classified as **hard** edges (no easy edges were found for this triangulation).

- **Hard Edge 1:** Vector = [1, 0, 0, 1, 0, 1, 1, 0, 1, 1, 0, 0]
- **Hard Edge 2:** Vector = [0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0]

Since both internal edges are hard, there are **two fugacity variables** η_1, η_2 in the refined index.

3 Neumann-Zagier Matrix

Row structure of $g_{NZ} \in \mathrm{Sp}(8, \mathbb{Q})$ ($2n \times 2n$ with $n = 4$, columns in block order $Z_1, Z_2, Z_3, Z_4, Z_1'', Z_2'', Z_3'', Z_4''$

$$\text{rows } 0,1 : \mu_0, \mu_1 \quad \text{rows } 2,3 : Q_1, Q_2 \quad \text{rows } 4,5 : \tfrac{\lambda_0}{2}, \tfrac{\lambda_1}{2} \quad \text{rows } 6,7 : \Gamma_1, \Gamma_2$$

g_{NZ} **Matrix:**

$$g_{NZ} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & -1 & -1 & 1 & 1 & -1 \\ -1 & 1 & -1 & 0 & 0 & 1 & 0 & -1 \\ -\frac{1}{2} & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \end{bmatrix}$$

Affine Shift ν**:**

$$\nu_X = \begin{bmatrix} -1 \\ -1 \\ -2 \\ 0 \end{bmatrix}, \qquad \nu_P = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

4 Inversion and Local Charges

Inverse Matrix g_{NZ}^{-1} **(via symplectic identity** $g^{-1} = [[D^T, -B^T], [-C^T, A^T]]$**):**

$$g_{NZ}^{-1} = \begin{bmatrix} 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & -\frac{1}{2} & 0 & 0 & -1 & 0 & -1 & -1 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & -1 & -1 & -1 & -1 \\ -1 & \frac{1}{2} & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & \frac{1}{2} & 0 & -1 & 0 & 0 & 1 & -1 \\ -1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & -\frac{1}{2} & 0 & 1 & 1 & 0 \\ 0 & -\frac{1}{2} & 1 & -\frac{1}{2} & 0 & 1 & 1 & -1 \end{bmatrix}$$

The master charge vector is

$$\kappa = \big[m_1,\ m_2,\ 0,\ 0,\ e_1,\ e_2,\ e_{\text{int},1},\ e_{\text{int},2}\big]^T,$$

where m_1, m_2 are cusp meridian charges, e_1, e_2 are cusp longitude/2 charges, and $e_{\text{int},1}, e_{\text{int},2} \in \frac{1}{2}\mathbb{Z}$ are summed over.

The local tetrahedron charges $\Delta = g_{NZ}^{-1}\kappa$:

$$\Delta_1: \quad \tilde{m}_1 = -\tfrac{m_2}{2} + e_{\text{int},2}, \qquad \tilde{e}_1 = m_1 + \tfrac{m_2}{2} + e_{\text{int},1} - e_{\text{int},2}$$

$$\Delta_2: \quad \tilde{m}_2 = m_1 - \tfrac{m_2}{2} - e_1 - e_{\text{int},1} - e_{\text{int},2}, \qquad \tilde{e}_2 = -m_1 + e_1 + e_{\text{int},1}$$

$$\Delta_3: \quad \tilde{m}_3 = \tfrac{m_2}{2} - e_1 - e_2 - e_{\text{int},1} - e_{\text{int},2}, \qquad \tilde{e}_3 = m_1 + e_2 + e_{\text{int},1}$$

$$\Delta_4: \quad \tilde{m}_4 = -m_1 + \tfrac{m_2}{2} + e_1 + e_{\text{int},2}, \qquad \tilde{e}_4 = -\tfrac{m_2}{2} + e_2 + e_{\text{int},1} - e_{\text{int},2}$$

The topological framing phase power $\phi = \sum_k \tilde{m}_k \cdot (\nu_P)_k - \sum_k \tilde{e}_k \cdot (\nu_X)_k$. Since $\nu_P = 0$ this reduces to:

$$\phi = -\big(\tilde{e}_1(-1) + \tilde{e}_2(-1) + \tilde{e}_3(-2) + \tilde{e}_4 \cdot 0\big) = \tilde{e}_1 + \tilde{e}_2 + 2\tilde{e}_3.$$

Substituting the local charges:

$$\phi = 2m_1 + \tfrac{m_2}{2} + e_1 + 2e_2 + 4e_{\text{int},1} - e_{\text{int},2}.$$

5 Final Summation Formula

The refined index is:

$$I_{5_1^2}(m_1, m_2, e_1, e_2) = \sum_{e_{\text{int}} \in \frac{1}{2}\mathbb{Z}^2} \left(\prod_{j=1}^{2} \eta_j^{2e_{\text{int},j}} \right) \cdot (-q^{1/2})^{\phi} \cdot \prod_{i=1}^{4} I_\Delta(\tilde{m}_i,\, \tilde{e}_i).$$

Evaluated output at $(m_1, m_2, e_1, e_2) = (0, 0, 0, 0)$ **(truncated at** q^4**):**

$$\begin{aligned}
I = 1 \\
&- 4q + q\,\eta_2^2 \\
&+ q^2\big(-2 + \eta_1^{-1} + \eta_1 + \eta_2^4\big) \\
&+ q^3\big(12 - 4\eta_2^2 + \eta_2^{-2} + \eta_1^{-1}\eta_2^{-2} + \eta_1\eta_2^{-2} + 2\eta_1^{-1} + 2\eta_1 + \eta_2^6\big) \\
&+ q^4\big(17 - 8\eta_2^2 + 4\eta_2^{-2} - \eta_1^{-1} + 2\eta_1^{-1}\eta_2^{-2} - \eta_1 + 2\eta_1\eta_2^{-2} + \eta_2^4 + \eta_2^8\big)
\end{aligned}$$

Raw key convention (as stored in `RefinedIndexResult`): a monomial $q^k\,\eta_1^a\,\eta_2^b$ is stored under key $(2k,\ 2a,\ 2b)$.

Remark on triangulation difference: The reference document `5_2_1_debug.tex` was generated from a different SnapPy triangulation of 5_1^2 (the pipeline found one easy and one hard edge there, yielding a 1×1 fugacity sector). The current pipeline's triangulation for `L5a1` has no easy edges, so the refined index naturally carries two fugacity variables η_1, η_2.